Comparison technique for semilinear elliptic problems on Riemannian Manifolds

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The Set Up

- Let (M^n, g) be a Riemannian manifold such that there exists $k \in \mathbb{R}$,
$$\mathrm{Ric}_g \geq k(n-1)g.$$
- Let $\Omega \subset M^n$ be a paracompact domain with $\mathscr{C}^2$ boundary.
- Let $f \in \mathrm{Lip}_{\mathrm{loc}}(\mathbb{R})$ be a locally Lipschitz function such that
$$f(x) > 0 \quad \text{and} \quad f(x) \geq nkx + f(0) \quad \text{for } x \in \mathscr{I}_f \subset \mathbb{R}_+,$$
with $f(0) > 0$ when $k \leq 0$.

Then, the suitable initial conditions for the comparison depending on f takes values in:
$$0 \in \mathscr{R}_f \subset [0, \bar{r}_k) \quad \text{and} \quad \mathscr{I}_f \subset \mathbb{R}_+ \text{ with } 0 \in \partial \mathscr{I}_f.$$
Then we consider the following **semilinear elliptic problem**,
$$\begin{cases} \Delta u + f(u) = 0 \text{ in } \Omega, \\ u > 0 \text{ in } \Omega, \\ u = 0 \text{ in } \partial\Omega. \end{cases} \qquad (☠)$$

Basis for the Comparison

Model manifold

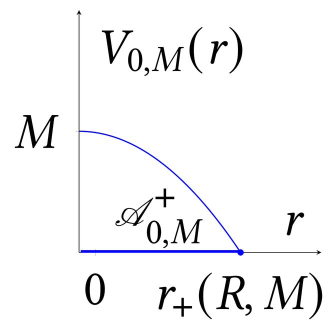
$$(M_k^n(F), g_k) := (I_k \times F,\ dr^2 + s_k(r)^2 g_F).$$

The Laplace–Beltrami operator on a model manifold,
$$\Delta = \partial_r^2 + (n-1)\cot_k(r)\,\partial_r + \frac{1}{s_k(r)^2}\Delta_F,$$

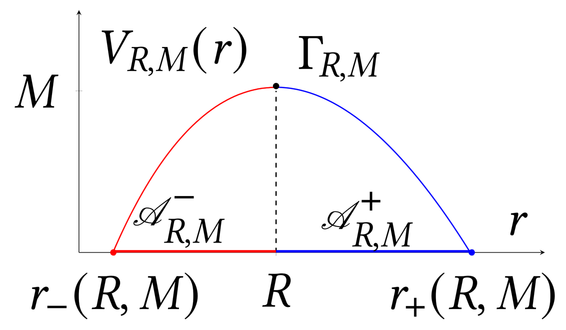
Model Solutions (*Radial solutions*)

$$v(p) = V(r(p))\ \forall p \in \mathscr{A} \Leftrightarrow \begin{cases} V''(r) + (n-1)\cot_k(r)V'(r) + f(V(r)) = 0, \\ V(R) = M, \quad V'(R) = 0. \end{cases} \qquad (☺)$$

For each initial data $(R, M) \in \mathscr{R}_f \times \mathscr{I}_f$, we have two types **model pairs**, pairs solutions-domains $(v_{R,M}, \mathscr{A}_{R,M})$:



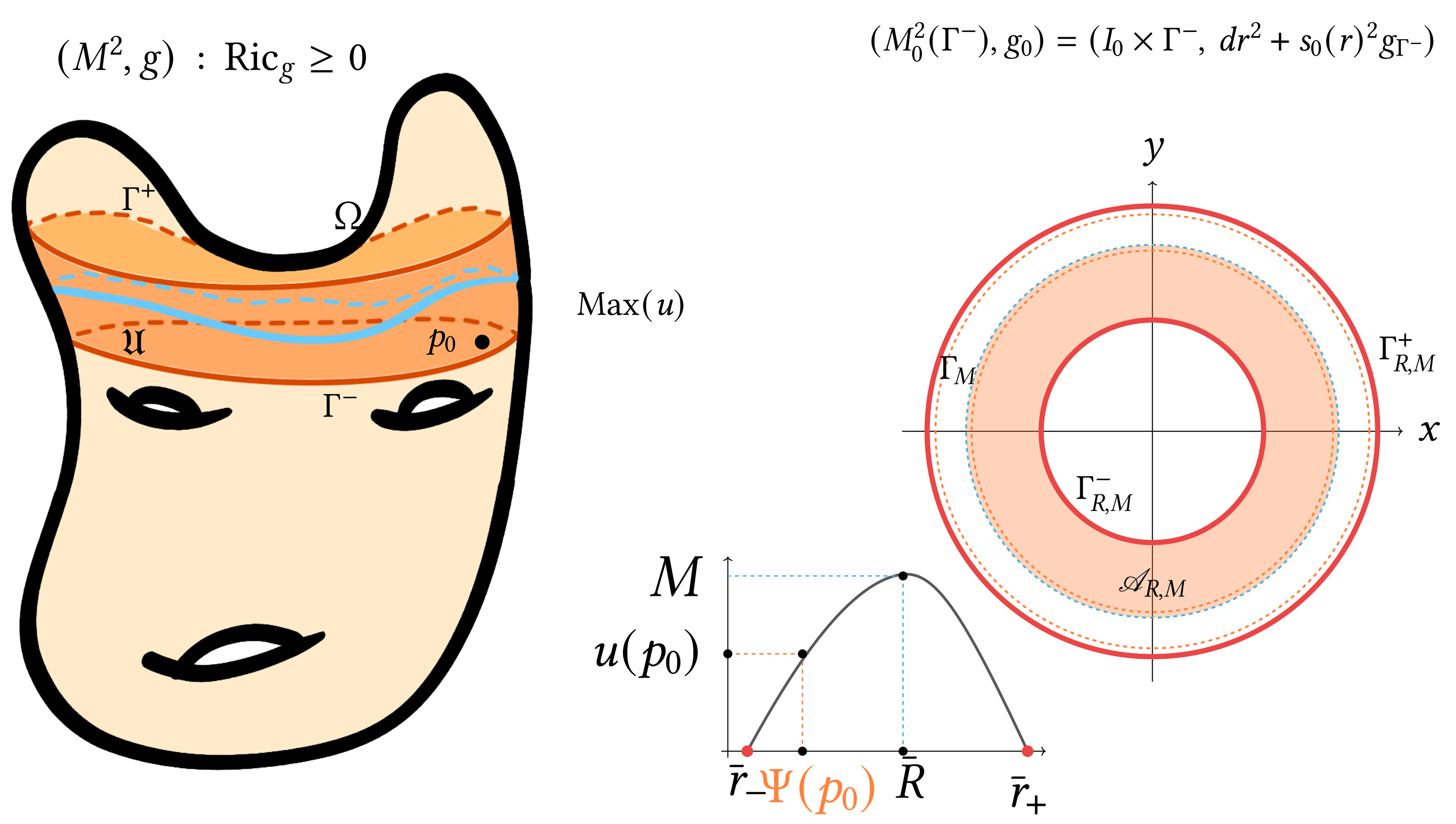
$\mathscr{A}_{0,M} = \{\, p \in M_k^n(F) : r(p) < r_+(0,M) \,\}$



$\mathscr{A}_{R,M} = \{\, p \in M_k^n(F) : r_-(R,M) < r(p) < r_+(R,M) \,\}$

Model $\overline{\tau}$-functions are defined as,
$$\overline{\tau}^{\pm}_M(R) := \frac{V'_{R,M}\big(r_\pm(R,M)\big)^2}{V'_{0,M}\big(r_+(0,M)\big)^2}$$

Comparison algorithm



1. **Choosing a connected component** We fix (Ω, u) and set $M := u_{\max}$. Let $\mathscr{U}$ be a connected component of $\Omega \setminus \mathrm{Max}(u)$.
$$\overline{\tau}(\Gamma) := \max_{p\in\Gamma} \frac{|\nabla u(p)|^2}{V'_{0,M}(r_+(0,M))^2},\ \overline{\tau}(\mathfrak{U}) := \max\{\overline{\tau}(\Gamma) : \Gamma \in \pi_0(\partial\Omega \cap \mathrm{cl}(\mathfrak{U}))\}$$
$$\&\ \overline{\tau}(\Omega) := \max\{\overline{\tau}(\mathfrak{U}) : \mathfrak{U} \in \pi_0(\Omega \setminus \mathrm{Max}(u))\}.$$
2. **Construction of the Model Manifold** Choose a boundary component $\Gamma_0 \in \pi_0(\partial\Omega \cap \mathrm{cl}(\mathscr{U}))$ such that $\overline{\tau}(\Gamma_0) = \overline{\tau}(\mathscr{U})$. We then work on the model manifold whose fiber is Γ_0 equipped with the metric induced by g on Γ_0, denoted by $\bar{g}$.
3. **Comparison criteria within the model model family** We say that a model pair in $M_k^n(\Gamma_0)$ $(\bar{\mathscr{A}}, \bar{v}) := (\mathscr{A}^\pm_{R,M}, v^\pm_{R,M}) \in \mathrm{Model}^\pm_M$ is:
   - A *comparison pair* for $(\mathfrak{U}, u)$ if $\overline{\tau}(\mathfrak{U}) \leq \overline{\tau}(\bar{\mathscr{A}}) = \overline{\tau}^\pm_M(R)$,
   - An *associated model pair* if $\overline{\tau}(\mathfrak{U}) = \overline{\tau}(\bar{\mathscr{A}}) = \overline{\tau}^\pm_M(R)$

   We fix $(\bar{\mathscr{A}}, \bar{v})$ a **comparison pair** with core radius $\bar{R}$ and $\bar{v}(\bar{r}_\pm) = 0$.
4. **Construction of the comparison tool** The *pseudo-radial function* associated to $(\mathfrak{U}, u)$ and $(\bar{\mathscr{A}}, \bar{v})$ is
$$\Psi : \mathfrak{U} \longrightarrow \chi^\pm(0, M) \subset [\bar{r}_-, \bar{r}_+], \qquad \Psi(p) := \chi^\pm(u(p)),$$
which will allow us to compare the geometry of level sets.

Comparison results

Gradient estimates

Let $f \in \mathscr{C}^1(\mathbb{R})$ satisfy
$$f'(x) \geq nk \qquad \text{for all } x \in \mathscr{I}_f. \qquad (\bigstar)$$
Then
$$|\nabla u|^2(p) \leq \bar{V}'(\Psi)^2(p) \qquad \text{for all } p \in \mathfrak{U}.$$
If equality holds at some point $p \in \mathfrak{U}$, then
$$\begin{cases} \mathrm{Ric}(\nabla u, \nabla u) = (n-1)k|\nabla u|^2 & \text{on } \mathfrak{U}, \\ (M^n, g) \cong (M_k^n(\Gamma_0), g_k) & \text{on } \mathfrak{U}, \\ (\mathfrak{U}, u) \equiv (\bar{\mathscr{A}}, \bar{v}). \end{cases}$$

Location of the hot-spots

Let $f \in \mathscr{C}^1(\mathbb{R})$ satisfy $(\bigstar)$ and
$$f(x) \leq f_k(x) = nkx + 1.$$
Assume that Ω is contained in a convex domain of M, $\overline{\tau}(\Omega) = \overline{\tau}(\mathfrak{U})$, and, if $k > 0$, $\bar{r}_+ \leq \bar{r}_k/2$.
Then, for all $p \in \mathrm{Max}(u)$,
$$\frac{\mathrm{dist}(p, \partial\Omega)}{\tan_k(r_\Omega)/n} \geq \frac{|\bar{r}_\pm - \bar{R}|}{\sqrt{\frac{2}{n}M + kM^2}}.$$
If equality holds at a point of Ω, then
$$\begin{cases} (M, g) \cong \mathbb{M}^n(k), \\ f(x) = nkx + 1, \\ (\Omega, u) \text{ is a ball solution of Serrin's problem.} \end{cases}$$

Isoperimetric type inequality

Let $f \in \mathscr{C}^m(\mathbb{R})$, with $m = \max\{n-2, 3\}$, satisfy $(\bigstar)$.
Assume that the $(n-1)$-dimensional part of $\mathrm{cl}(\mathfrak{U}) \cap \mathrm{Max}(u)$, denoted by Γ_M, is a possibly disconnected Lipschitz hypersurface.
Then
$$\frac{\mathscr{H}^n(\mathfrak{U})}{\mathscr{H}^{n-1}(\Gamma_M)} \geq \frac{\mathscr{H}^n_k(\bar{\mathscr{A}})}{\mathscr{H}^{n-1}_k(\bar{\Gamma}_M)}.$$
Equality holds if and only if
$$(\mathfrak{U}, u) \equiv (\bar{\mathscr{A}}, \bar{v}).$$

Notation

- $I_k := [0, \bar{r}_k)$, where $\bar{r}_k := +\infty$ if $k \leq 0$ and $\bar{r}_k := \pi/\sqrt{k}$ if $k > 0$.
- $s_k(r) := \begin{cases} \frac{\sinh(\sqrt{-k}r)}{\sqrt{-k}} & \text{if } k < 0 \\ r & \text{if } k = 0, \\ \frac{\sin(\sqrt{k}r)}{\sqrt{k}} & \text{if } k > 0 \end{cases} \quad \cot_k(r) := \frac{s_k'(r)}{s_k(r)} \quad \text{and} \quad \tan_k(r) = \frac{s_k(r)}{s_k'(r)}.$
- $r_\Omega = \max\{\mathrm{dist}(p, \partial\Omega) : p \in \Omega\}$

Bibliography

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